

## Training Configuration and Summary:

Dataset used: rssi\_odom\_dataset\_12apr25\_ajust.parquet

model\_type: LSTM  
group\_sec: 40  
timesteps\_orig: 60  
batch\_size: 256  
min\_units: 16  
max\_units: 1024  
units\_step: 16  
dropout\_min: 0.2  
dropout\_max: 0.3  
dropout\_step: 0.1  
activation\_functions: ['tanh', 'relu', 'elu']  
optimizers: ['adam', 'rmsprop', 'Nadam']  
min\_layers: 1  
max\_layers: 2  
max\_epochs: 20  
hyperband\_iterations: 3  
group\_msec: 40  
samples\_per\_sec: 25  
forecast\_horizon\_sec: 4  
forecast\_steps: 100  
timesteps: 60

### LSTM Multi-Step Training Report

Report Generated: 2025-05-21 21:10:23

Training Start: 2025-05-21\_14-00-14

Training End: 2025-05-21\_15-00-58

Training Duration: 60.73 minutes

Predictions Start: 2025-05-21\_15-00-58

Predictions End: 2025-05-21\_15-04-13

Predictions Duration: 3.25 minutes

#### Best Hyperparameters:

Number of Layers: 1

Activation: tanh

Optimizer: rmsprop

Dropout Rate: 0.20

MAE: 1.9926

MSE: 5.9348

RMSE: 2.4361

R2 Score: 0.6550

#### Multi-Step Forecast:

Forecast Steps: 100

MAE: 0.0356

MSE: 0.0018

RMSE: 0.0429

R2 Score: 0.8076

Dataset size before down-sample: 85,269 rows  
Dataset size after down-sample: 12,829 rows

First 5 timestamps BEFORE:

1744397085760  
1744397085770  
1744397085789  
1744397085801  
1744397085817

...

Last 5 timestamps BEFORE:

1744464971366  
1744464971377  
1744464971390  
1744464971402  
1744464971414

First 5 timestamps AFTER:

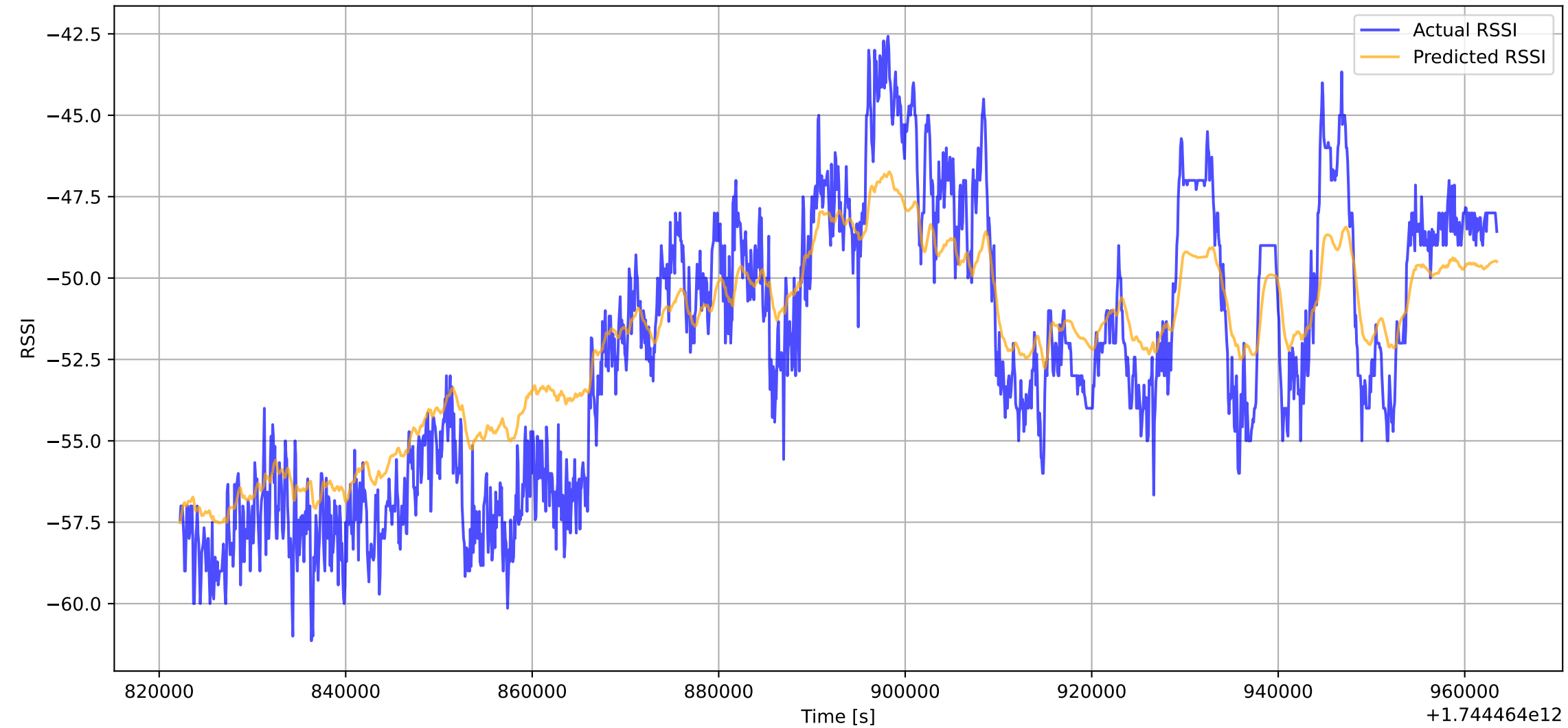
1744397085760  
1744397085841  
1744397086014  
1744397086103  
1744397086172

...

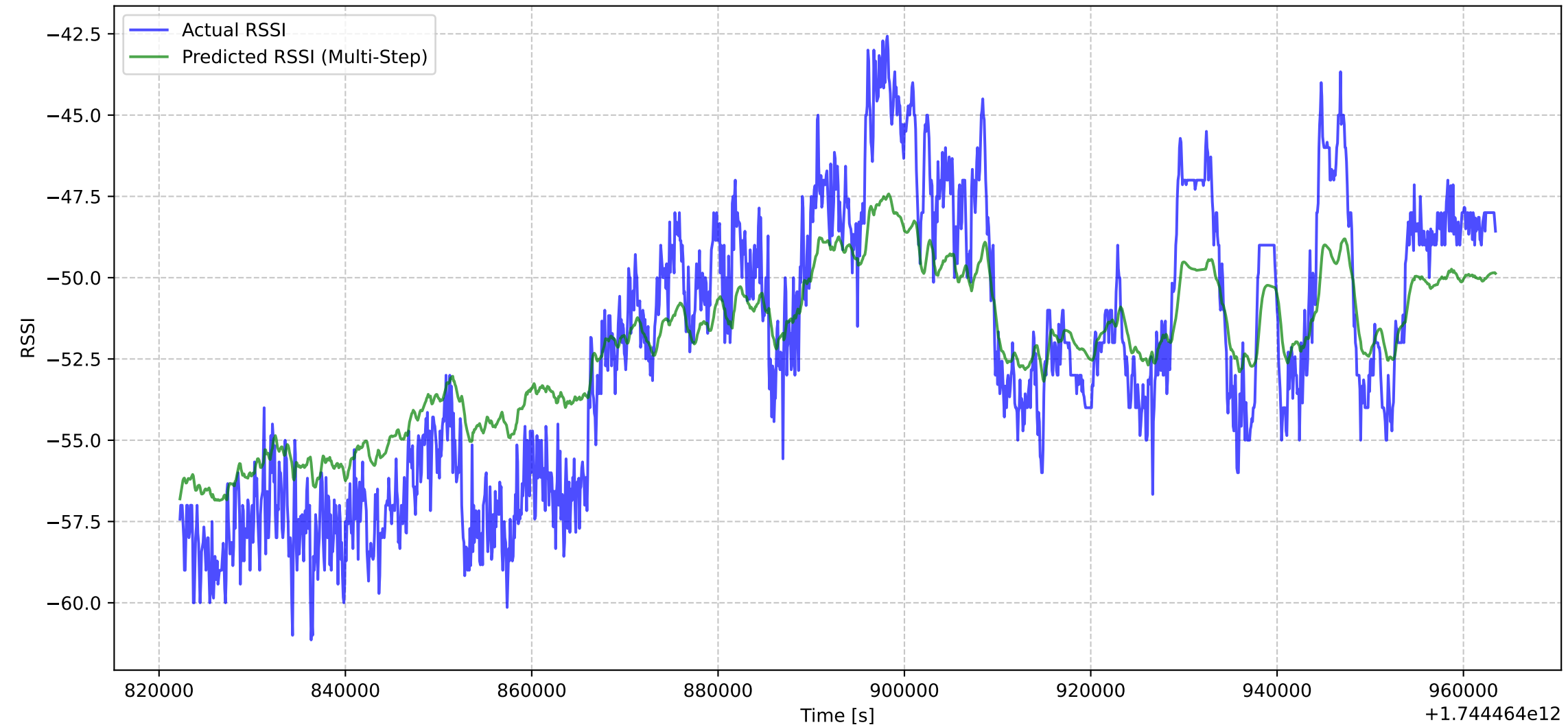
Last 5 timestamps AFTER:

1744464971041  
1744464971124  
1744464971209  
1744464971281  
1744464971366

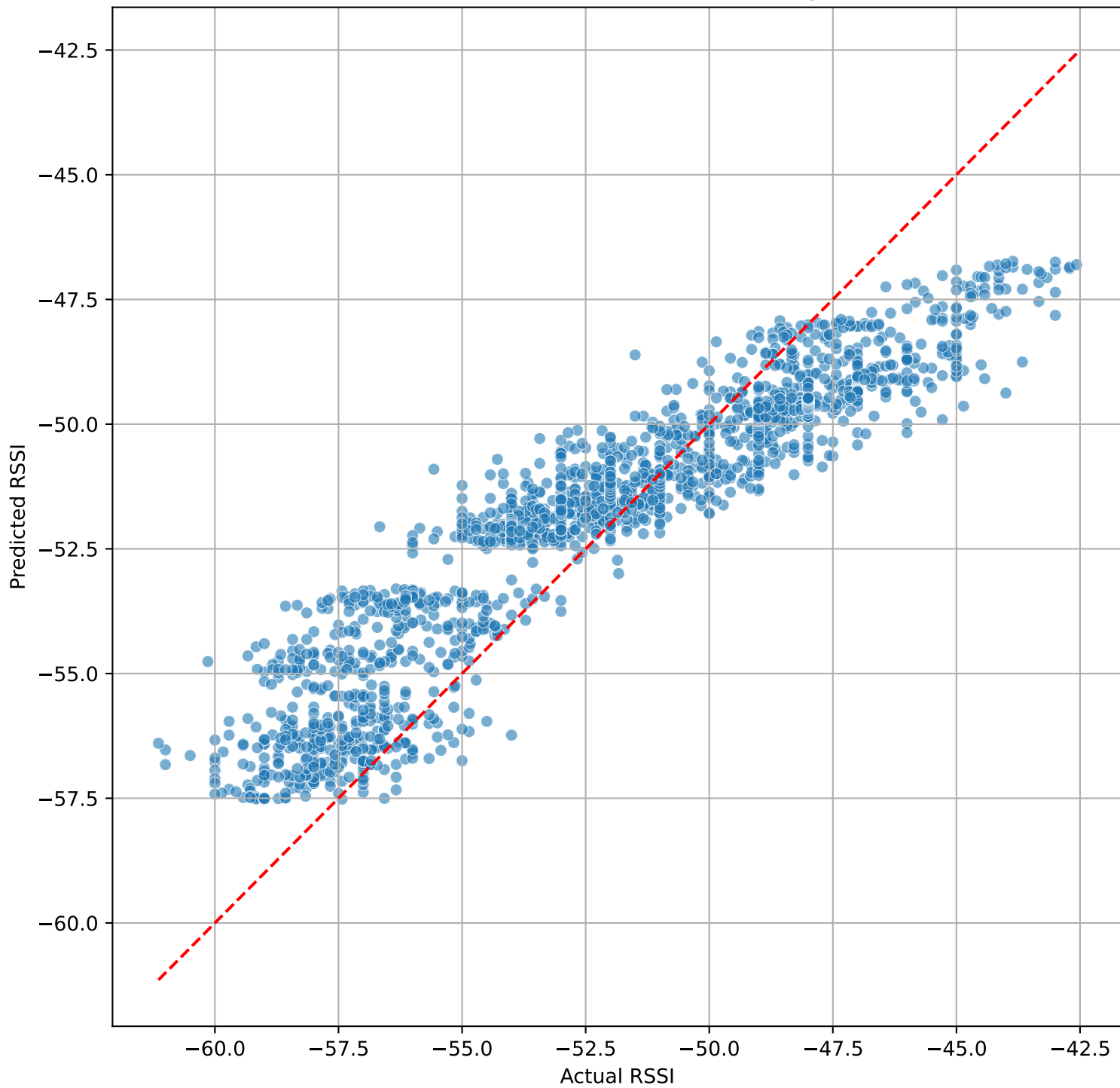
Actual vs Predicted RSSI (First Step)



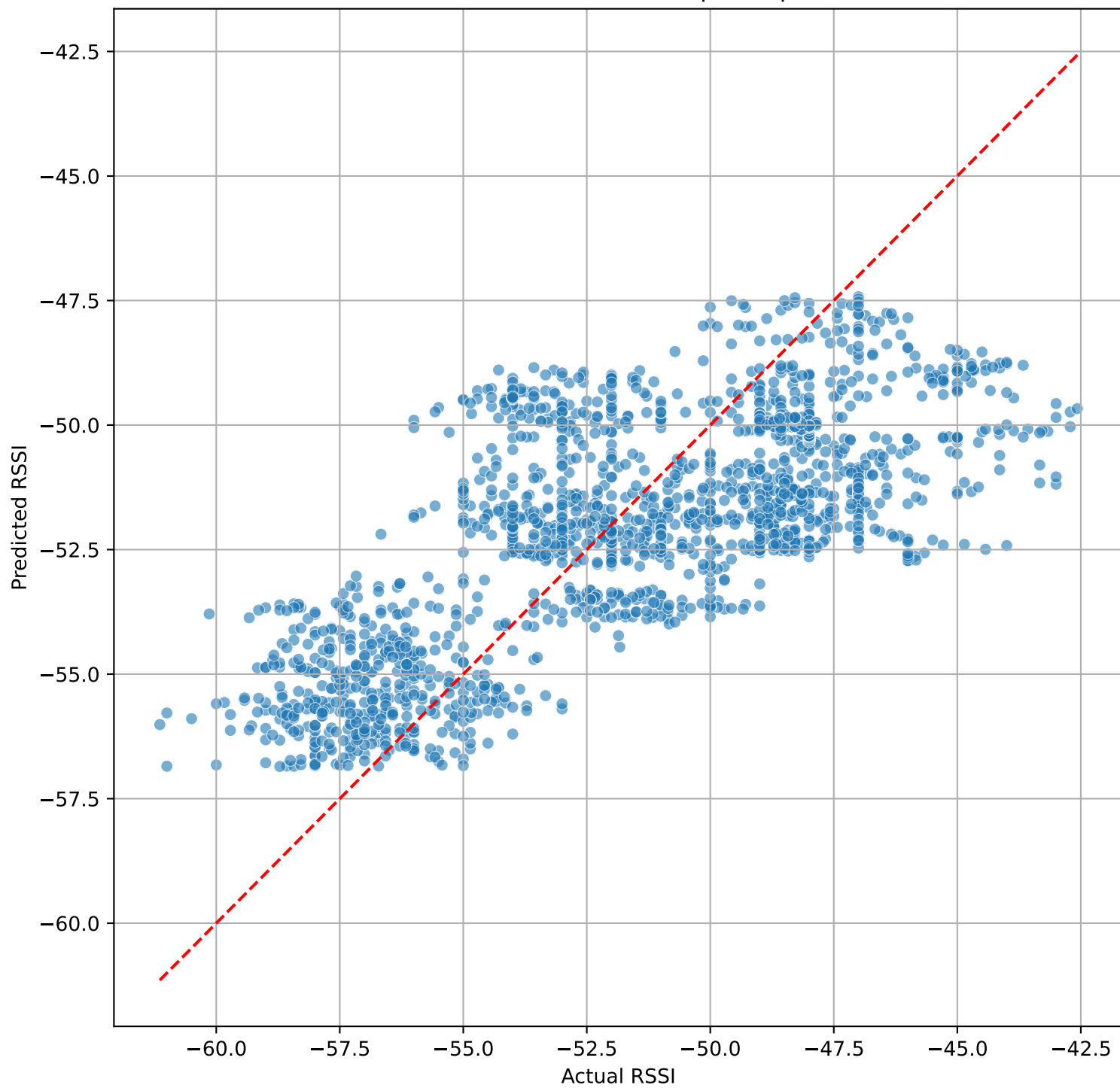
Actual vs Predicted RSSI (Multi-Step (step 100 $\approx$  4.0 s))



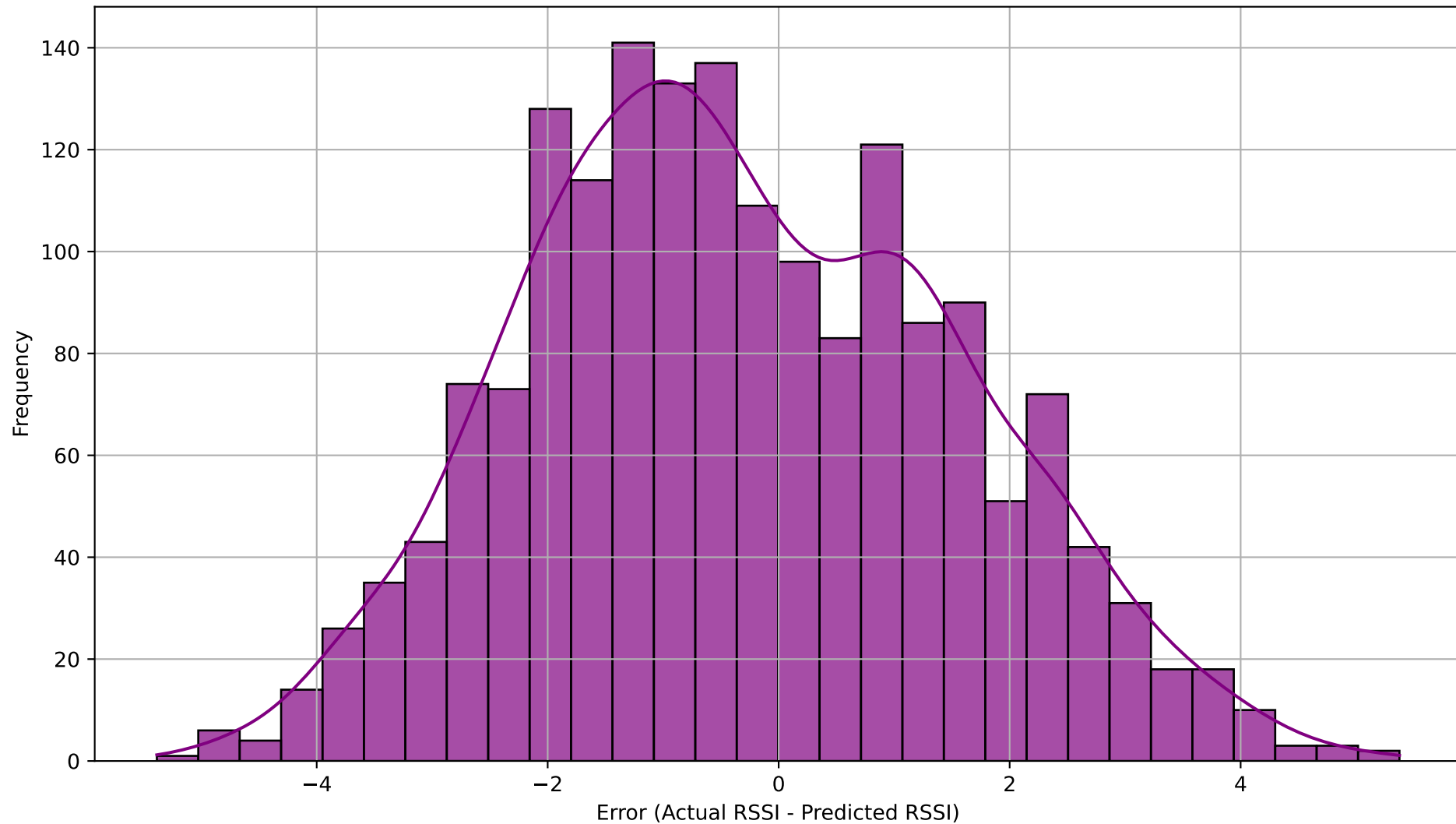
Actual vs Predicted RSSI (First Step)



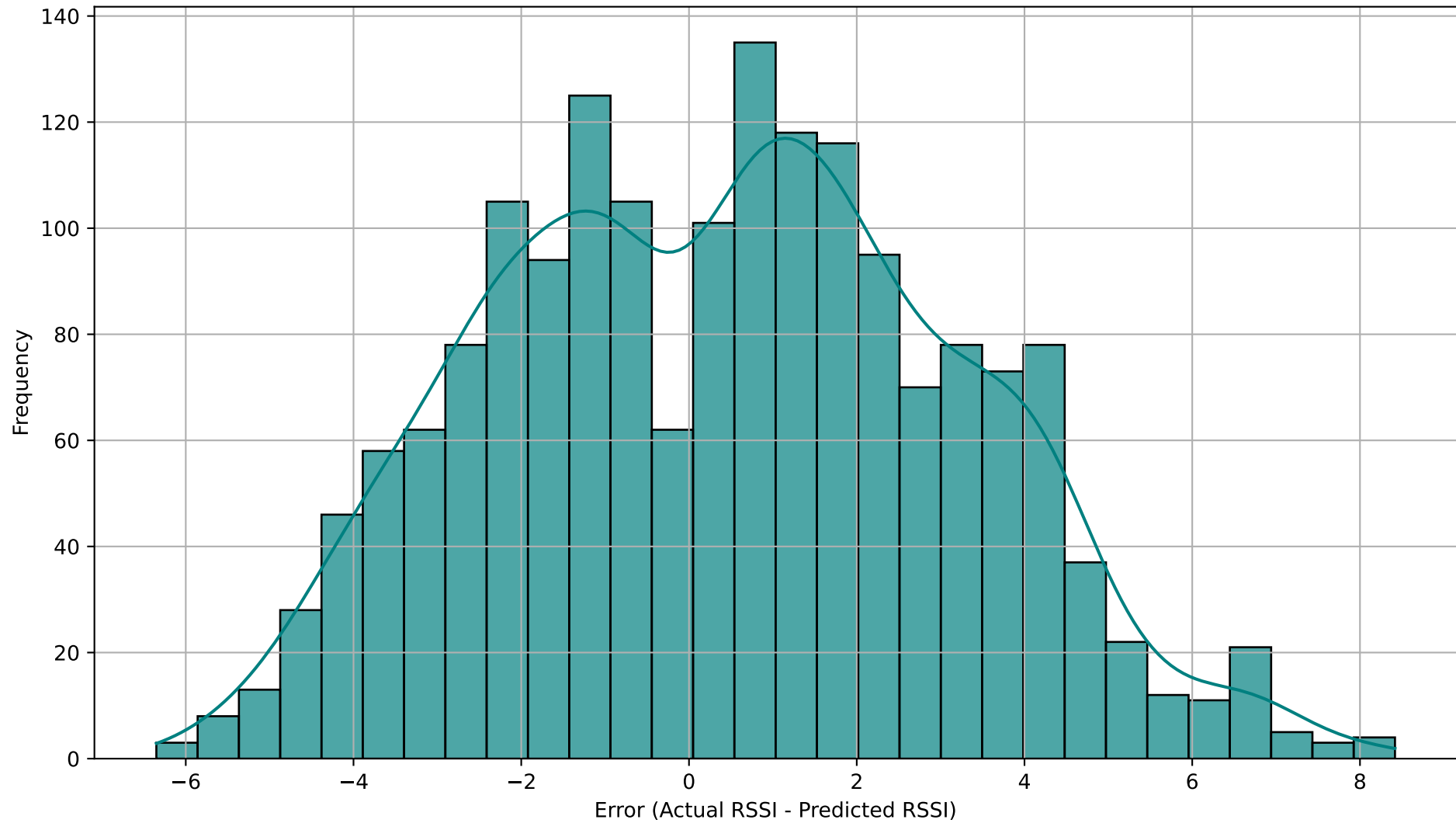
Actual vs Predicted RSSI (Last Step - step 100  $\approx$  4.0 s)



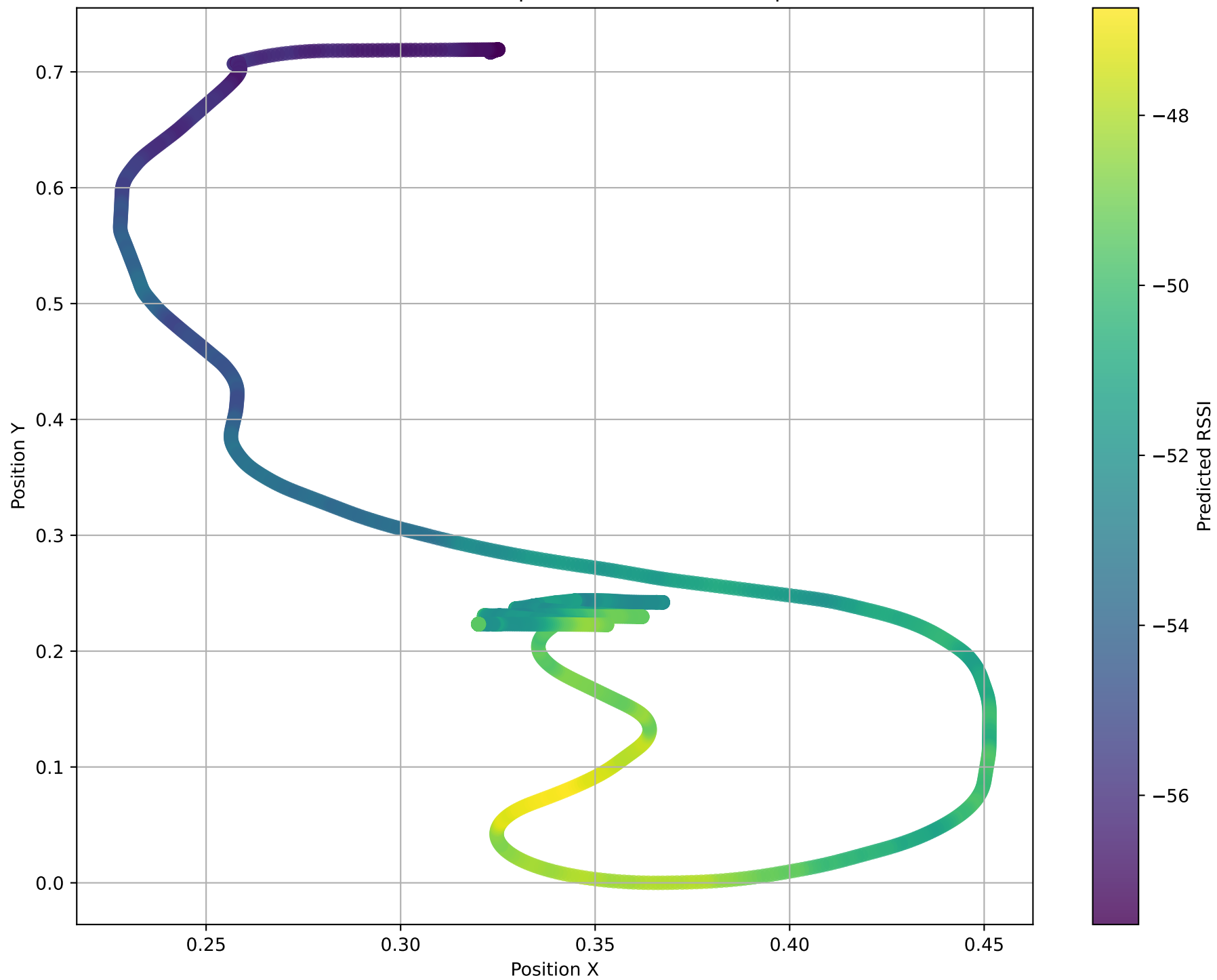
Distribution of Prediction Errors (First Step)



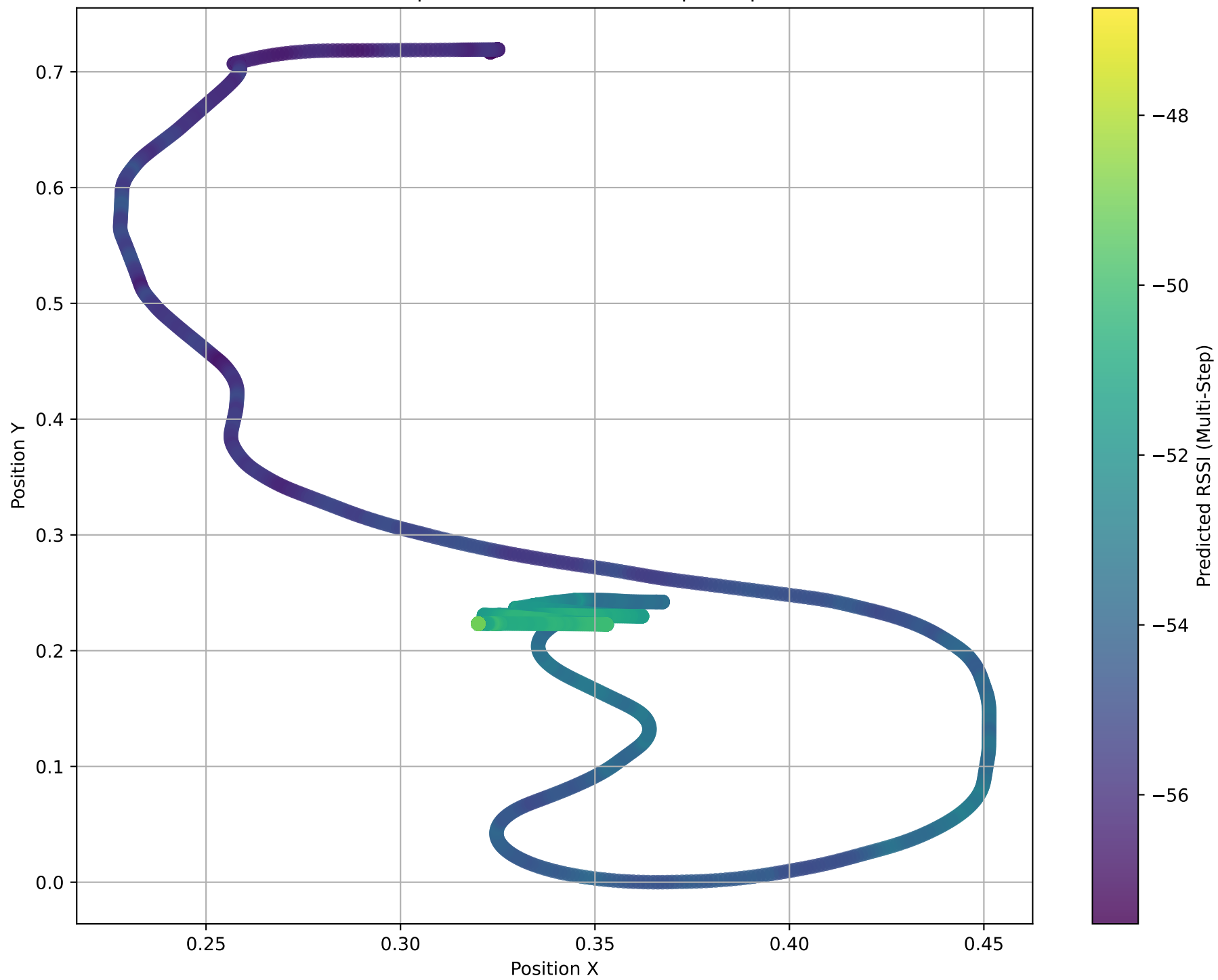
Distribution of Prediction Errors (Last Step - step 100  $\approx$  4.0 s)



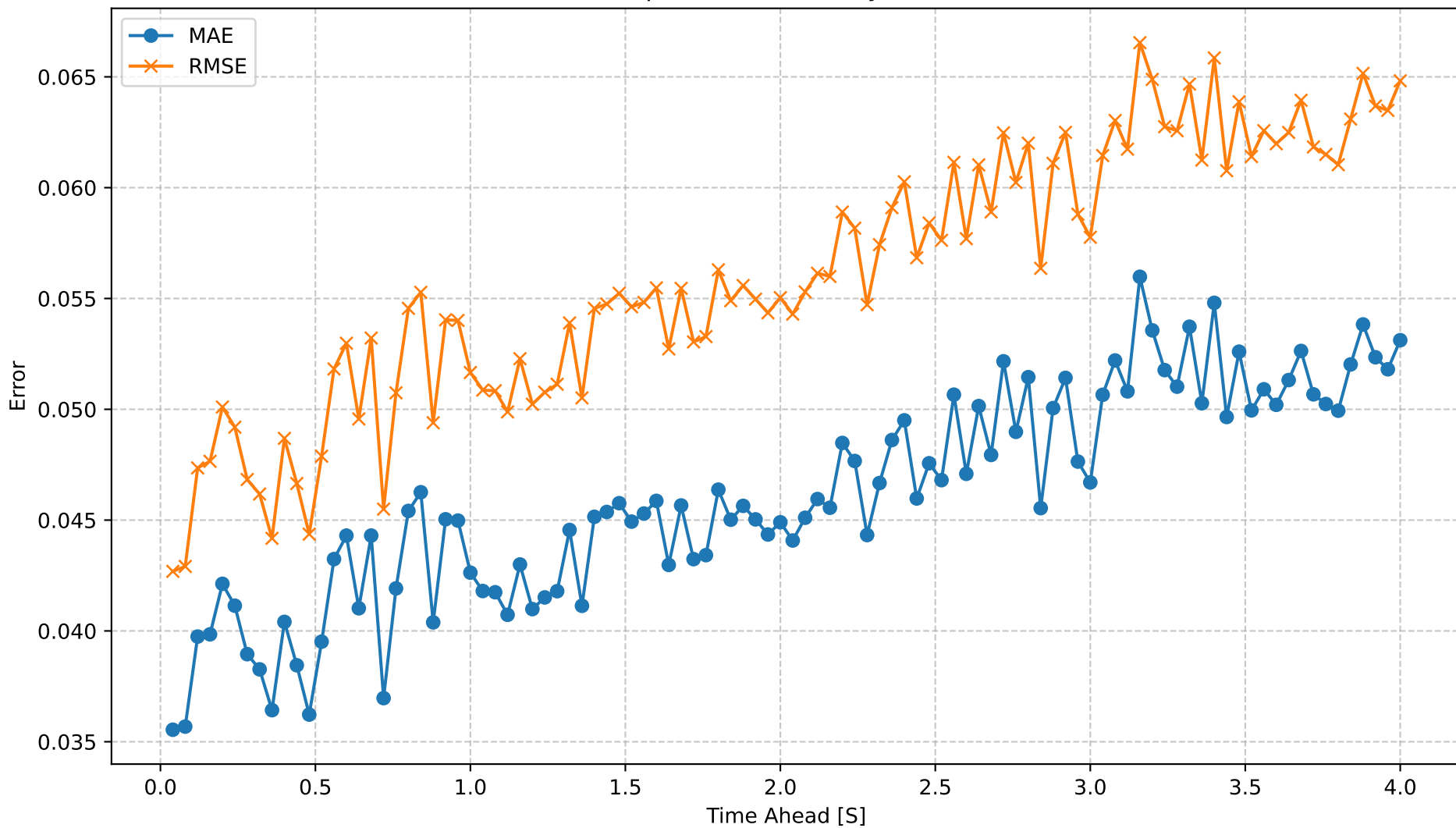
Predicted RSSI in Spatial Context (First Step)



Predicted RSSI in Spatial Context (Multi-Step - step 100 $\approx$  4.0 s)



Multi-Step Forecast Error by Time Ahead



## Resumo Automático

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Dataset: rssi\_odom\_dataset\_12apr25\_ajust.parquet  
Modelo: LSTM  
Forecast steps: 100

### MAE / RMSE (dB)

- 1º passo : 0.04 / 0.04
- Mediana : 0.05 / 0.06
- 100º passo : 0.05 / 0.06

Crescimento do erro: 49.5% → Moderado (erro cresce 20-50 %)

3 piores passos (MAE): 79 (0.06), 85 (0.05), 97 (0.05)

### Recomendação:

Use as previsões até a metade do horizonte; após isso a incerteza cresce consideravelmente.